

Careers in AI & Transportation



15 STEM professions, salary ranges, and education pathways

HIGH
SCHOOL

CAREER
GUIDE

Salary figures are approximate U.S. national averages and vary by location, experience, and employer. Use this guide as a starting point for your own research.

Career	What They Do	Typical Education	Salary
Machine Learning Engineer	Builds and trains AI models that learn from data.	Bachelor's in CS or related field	\$110K-\$165K
Data Scientist	Analyzes large datasets to find patterns and inform decisions.	Bachelor's/Master's in Data Science or CS	\$100K-\$150K
Traffic Systems Engineer	Designs and optimizes road and signal systems for cities.	Bachelor's in Civil/Transportation Eng.	\$75K-\$110K
Cybersecurity Analyst	Protects networks and systems from cyberattacks.	Bachelor's in CS/Cybersecurity + certs	\$80K-\$120K
Robotics Engineer	Designs and programs robots and autonomous systems.	Bachelor's in Robotics/Mech./EE	\$85K-\$130K
AI Ethics Researcher	Studies fairness, bias, and responsible AI policy.	Master's/PhD in CS, Philosophy, or Policy	\$90K-\$140K
Software Developer	Builds the applications and systems that power technology.	Bachelor's in CS or coding bootcamp	\$80K-\$130K
Urban Planner	Plans how cities grow, including transportation networks.	Master's in Urban/City Planning	\$60K-\$95K
Network Security Engineer	Designs secure infrastructure for organizations.	Bachelor's in CS/IT + security certs	\$95K-\$140K
Autonomous Vehicle Engineer	Develops self-driving car perception/control systems.	Bachelor's/Master's in EE, CS, Robotics	\$110K-\$160K
Data Engineer	Builds the pipelines that move and organize big data.	Bachelor's in CS or Data Engineering	\$95K-\$140K
UX Researcher (AI Products)	Studies how people interact with AI-powered products.	Bachelor's/Master's in HCI or Psychology	\$85K-\$125K
IT Support / Sys Admin	Maintains and secures an organization's computer systems.	Associate's degree or certifications	\$50K-\$80K
Smart Infra. Technician	Installs and maintains sensors and smart city hardware.	Associate's degree or trade cert.	\$45K-\$70K

Career	What They Do	Typical Education	Salary
Computer Vision Engineer	Builds systems that let computers 'see' and interpret images.	Bachelor's/Master's in CS or EE	\$110K-\$160K



Education Pathways

How to get from here to a STEM career

- **High School:** Take AP Computer Science, AP Statistics, or Engineering electives if available. Join robotics, coding, or cybersecurity clubs.
- **Associate's Degree (2 years):** A fast path into IT support, networking, or infrastructure technician roles, often with industry certifications like CompTIA.
- **Bachelor's Degree (4 years):** Required for most engineering, data science, and software roles. Common majors: Computer Science, Computer Engineering, Civil/Transportation Engineering.
- **Master's/PhD:** Useful for research-focused roles like AI Ethics, advanced Machine Learning, or academic positions.
- **Certifications:** Many cybersecurity and cloud roles value certifications (CompTIA Security+, AWS Certified, Google Data Analytics) alongside or instead of a degree.

How to Start Exploring Now

Take Action This Week

Try free online courses (Khan Academy, Code.org, freeCodeCamp) · Join or start a robotics, coding, or cybersecurity club · Look for free summer programs at local universities · Ask to shadow a STEM professional · Use the AI Traffic Simulation on this website to explore transportation engineering hands-on.

